

# SURAJ PATEL MUTHE GOWDA

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## EDUCATION

### Northeastern University

Master's in Computer Engineering (Machine Learning and Computer Vision) [GPA 4.0/4.0]

Boston, MA

Expected Dec '26

**Relevant Coursework:** High-Performance Computing, Machine Learning, LLM Agents, MLOps

### PES University

Bachelor's in Electrical and Electronics Engineering (Embedded Systems) [GPA 3.8/4.0]

Bengaluru, India

May '22

**Relevant Coursework:** Computer Architecture, Data Structures & Algorithms, Operating Systems

## TECHNICAL SKILLS

**Languages:** Python, C++, Java, SQL, JavaScript/TypeScript, Bash

**Machine Learning:** PyTorch, TensorFlow, scikit-learn, Transformers, CNNs, LSTMs, NLP, Multimodal Learning, Hugging Face

**Generative AI:** LLMs, AI Agents, RAG, Prompt/Context Engineering, Agentic Evaluation, LangChain, Embeddings

**ML Systems & Services:** Model Evaluation, MLflow, FastAPI, REST APIs, WebSockets, ONNX, Quantization, React

**Cloud & Infrastructure:** AWS (ECS Fargate, S3, SQS, EventBridge), Docker, Kubernetes, CI/CD, Git, Linux/Unix

## WORK EXPERIENCE

### Amazon

Seattle, WA

Software Development Engineer (AI) Intern

Jun - Aug '26

- Built a **scalable service** on **AWS** (ECS Fargate, SQS, EventBridge, S3) to **analyze, attribute, and optimize 15K+ token system prompts** across production AI agents, running controlled experiments to identify **load-bearing and filler** prompt sections
- Developed a **statistical decision framework** using **OLS regression, Wilcoxon tests, bootstrap confidence intervals, and Bonferroni correction** to quantify prompt-section impact across **7 independent quality dimensions**
- Designed a **held-out non-inferiority validation pipeline** using **section-level ablation-driven removal** to counter interaction effects, enabling up to **12% prompt-token reduction** with no significant **response quality** loss

### Apnimed Inc.

Cambridge, MA

Machine Learning Research Co-op

Jul - Dec '25

- Built a **CNN-BiLSTM time-series classification pipeline** across **200K+ labeled wearable samples**, integrating SpO<sub>2</sub>, pulse rate, and movement signals to improve 4-class sleep-stage agreement from **0.60 to 0.80 Cohen's kappa**
- Engineered **data extraction, preprocessing, quality filtering, and class-balanced sampling pipelines** for **200K+ samples**, achieving **94.52% sensitivity** with a **5.48% false-negative rate** in Sleep/Wake classification
- Developed a **configurable evaluation and monitoring framework** with per-class metrics, **KL/JS distribution-shift detection**, and MLflow tracking, achieving **0.8132 AUROC** and **62.37% wake specificity**

### Samsung Research and Development Institute

Bengaluru, India

Senior Machine Learning Engineer

Jan '22 - Aug '24

- Developed a **real-time multimodal ML system** fusing call transcripts, OCR text, and sequential features with Transformer/ViViT encoders, achieving **94.44% accuracy** and a **16.48 point improvement** over transcript-only detection
- Optimized and productionized the model for **on-device inference** on Samsung S24 using **Samsung Neural SDK** and **16-bit Post-Training Quantization**, achieving **250 ms latency** while reducing model size by **40%** and runtime memory by **38%**
- Developed and deployed a **BERT-based threat classification pipeline** on adversarial security datasets, integrating fine-tuning, evaluation, and production monitoring to improve detection **accuracy by 20%**

## PROJECTS

### VideoCommentary AI - Vision-Language Pipeline for Video Understanding

Jan '26 - May '26

- Built a **vision-language generation model** combining a **BLIP-2** vision encoder with a custom **Transformer decoder** to generate context-aware natural-language commentary from temporally sampled video frames
- Developed an **end-to-end training pipeline** over annotated video-text pairs using temporal frame aggregation and **mixed-precision computation** to improve training efficiency on Apple Silicon

### Real-Time Courtroom Language Accessibility System

Jan '26 - Apr '26

- Built a **real-time multimodal inference system** to improve courtroom accessibility, integrating **speech-to-text, machine translation, text-to-speech, and ASL generation** into a unified pipeline targeting **~2s end-to-end latency**
- Developed a **FastAPI/WebSocket streaming backend** to orchestrate audio ingestion, model inference, and multimodal output delivery, incorporating **human-in-the-loop confidence handling** to route low-confidence predictions for validation

## PUBLICATIONS & CONFERENCE PRESENTATIONS

### Deep Learning Optimization of Sleep Stage Classification | ATS '26 | [Link](#)

May '26

- Built the **deep learning sleep-stage classification pipeline** using wearable-derived SpO<sub>2</sub>, pulse rate, and movement signals, improving 4-class agreement from **0.60 to 0.80 Cohen's kappa** across **200K+ labeled samples**

### AI-Driven Architectures for Real-Time Decision-Making in Autonomous Vehicles | IEEE ICAD '25 | [Link](#)

Jun '25

- Developed an **AI-driven architecture for real-time autonomous-vehicle decision-making**, integrating intelligent perception and decision frameworks for dynamic driving environments

### Multimodal Strategy to Defend Mobile Devices Against Vishing Attacks | ACM MobiCom '24 | [Link](#)

Dec '24

- Developed a **multimodal vishing-detection approach** combining conversational and visual signals to identify social-engineering attacks on mobile devices in real time